

Other Services Employment Impact



We are working on Moody's dataset with multiple features and responses for each market. It is quarterly data that means, that every quarter, new record come for each market. It is real estate dataset.

I am currently working with feature name `Other_Services_Employment_Extended` and response vacancy. Help me understand how it worked

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`Other_Services_Employment_Extended` is actually a very interesting macro feature for real estate vacancy modeling because it captures the health of the **local service economy outside the major sectors** like manufacturing, finance, tech, retail, or government.

In Moody's and similar macro datasets, this variable usually includes employment in categories such as:

- repair & maintenance services
- salons/personal care
- religious organizations
- laundry services
- parking services
- civic organizations
- private household services
- small local service businesses
- miscellaneous neighborhood services

It represents the "local economic activity density" of a market.

Why this matters for vacancy

The relationship depends heavily on property type.

1. Apartment Vacancy

This feature often has a **negative relationship** with apartment vacancy.

Meaning:

- Higher `Other_Services_Employment_Extended`
 - more local workers
 - stronger neighborhood economy

- more renters
- lower apartment vacancy

Why?

These jobs are usually:

- local
- labor intensive
- less remote-work friendly
- geographically sticky

A city with strong miscellaneous service employment means:

- restaurants operate
- local commerce survives
- neighborhood demand remains healthy

That supports apartment occupancy.

Example

Suppose:

Market	Other Services Employment	Apartment Vacancy
Austin	High	Low
Detroit	Weak	High

Austin may have:

- active urban consumption
- lifestyle economy
- local services growth

This creates housing demand.

2. Office Vacancy

This feature may have:

- weak relationship
- OR
- lagged indirect relationship

Because office demand is more driven by:

- finance
- tech
- professional services
- corporate hiring

However, if “other services employment” collapses badly, it may indicate:

- weak downtown ecosystem
- declining foot traffic
- reduced urban vitality

That can indirectly increase office vacancy later.

3. Retail Vacancy

This is where the feature becomes VERY useful.

Retail properties depend heavily on:

- neighborhood activity
- consumer movement
- local employment ecosystem

If other service employment rises:

- salons rent retail space
- repair shops occupy strip centers
- local businesses survive longer

So:

- retail occupancy improves
- vacancy declines

This feature can be a hidden proxy for:

- small business formation
 - neighborhood commercial health
-

4. Industrial Vacancy

Usually weaker relationship.

Industrial depends more on:

- logistics

- manufacturing
- imports/exports
- warehousing demand

But in some local economies:

- service growth indirectly boosts local distribution demand.
-

Important Statistical Behavior

This variable often behaves like a:

“Late-cycle local economy indicator”

Why?

Other services employment is:

- cyclical
- discretionary
- sensitive to local spending

During recession:

- consumers cut discretionary services
- small businesses close
- employment falls quickly

So the variable can act as an early warning signal for:

- weakening occupancy
 - rising future vacancy
-

Very Important Real Estate Interpretation

Institutional CRE firms often use variables like this to measure:

- neighborhood economic resilience
- urban vibrancy
- consumer activity depth
- local labor stickiness

This is especially valuable in:

- multifamily

- neighborhood retail
- mixed-use assets

How It Usually Appears in Models

A. Growth Rate Features

Instead of raw level:

$$YoY\ Growth = \frac{Employment_t - Employment_{t-4}}{Employment_{t-4}}$$

This captures economic momentum better.

$$YoY\ Growth = \frac{Employment_t - Employment_{t-4}}{Employment_{t-4}}$$

B. Employment Shock Features

Real estate reacts strongly to sudden declines.

Example:

$$Shock_t = Employment_t - RollingMean_{8Q}$$

Negative shock:

- signals local stress
- vacancy may rise 2-4 quarters later

C. Share of Total Employment

Very powerful feature:

$$OtherServicesShare = \frac{OtherServicesEmployment}{TotalEmployment}$$

$$OtherServicesShare = \frac{OtherServicesEmployment}{TotalEmployment}$$

This measures:

- economic composition
- service economy dependence

Markets overly dependent on low-margin local services may become fragile during downturns.

Why Extended Version Matters

The word **“Extended”** usually means:

- broader NAICS coverage
- revised historical continuity
- smoother benchmark adjustments

Moody’s often creates extended series to:

- backfill history
- reduce discontinuities
- improve forecasting stability

This is important for quarterly forecasting models because:

- standard employment series may have revisions/breaks
- extended versions improve time-series consistency

What Top CRE Firms Usually Do With It

Top institutional real estate firms rarely use this feature alone.

Instead they combine it with:

Combine With	Why
Retail sales	consumer strength
Personal income	spending power
Population growth	demand base
Small business formation	leasing demand
Transit mobility	urban activity
Wage growth	affordability

Most Important Practical Insight

This variable is usually NOT a direct driver of vacancy.

It is more of a:

“local economic ecosystem health indicator”

That’s why it becomes powerful in:

- lagged features
- interaction features

- regime models
- downturn prediction

Realistic Relationship by Property Type

Property Type	Expected Impact on Vacancy
Apartment	Strong negative
Retail	Strong negative
Office	Mild negative
Industrial	Weak/indirect

Very Strong Advanced Feature Ideas

1. Service Employment Stress

$$Stress = \frac{Employment_t}{RollingMax_{12Q}}$$

Low value indicates local economy deterioration.

$$Stress = \frac{Employment_t}{RollingMax_{12Q}}$$

2. Local Service Recovery Speed

After recession:

$$Recovery = Employment_t - Employment_{recession\ low}$$

Markets recovering faster usually see:

- faster leasing recovery
- lower vacancy persistence

3. Interaction With Consumer Income

Very realistic institutional feature:

$$Interaction = OtherServicesEmploymentGrowth \times IncomeGrowth$$

$$Interaction = OtherServicesEmploymentGrowth \times IncomeGrowth$$

Because:

- employment without income growth may be low-quality growth
- income + services growth together indicate strong local demand

Final Intuition

Think of `Other_Services_Employment_Extended` as:

“How alive and economically active the local neighborhood ecosystem is.”

If this ecosystem strengthens:

- people stay employed locally
- neighborhoods remain active
- small businesses survive
- retail traffic improves
- apartment demand stabilizes

And eventually:

- vacancy tends to decline.



Help with out of box features engineering for this column.

For `Other_Services_Employment_Extended`, the best feature engineering is usually **not simple lags or growth rates**.

Top CRE and macro forecasting teams try to transform it into signals about:

- local economic resilience
- neighborhood commercial intensity
- labor fragility
- recovery persistence
- service-sector dependency
- urban vibrancy cycles

Below are more advanced and realistic feature ideas specifically designed for quarterly real estate vacancy modeling.

1. Service Employment Momentum Acceleration

Not just growth — growth acceleration.

$$Acceleration_t = YoY Growth_t - YoY Growth_{t-1}$$

$$Acceleration_t = YoY Growth_t - YoY Growth_{t-1}$$

Why useful:

- detects turning points early
- vacancy often reacts to second derivative changes
- leasing markets care about momentum shifts

Interpretation:

- positive acceleration → improving local ecosystem
- negative acceleration → hidden slowdown before vacancy rises

2. Local Service Economy Regime Score

Create market states.

Example:

Regime	Condition
Expansion	growth > 4%
Stable	1-4%
Weakening	-2% to 1%
Stress	< -2%

Then encode regime.

Why powerful:

- vacancy behaves nonlinearly
- recessionary service collapse impacts vacancy disproportionately

This works very well with:

- XGBoost
- LightGBM
- CatBoost

3. Service Employment Relative to Population

$$ServiceDensity = \frac{OtherServicesEmployment}{Population}$$

$$ServiceDensity = \frac{OtherServicesEmployment}{Population}$$

This measures:

- local service intensity

- urban commercial depth
- neighborhood consumption infrastructure

Very important for:

- retail
- multifamily
- mixed-use assets

Markets with high service density often have:

- lower structural vacancy
- stronger tenant retention

4. Employment Efficiency Feature

$$Efficiency = \frac{OtherServicesEmployment}{RetailSquareFeet}$$

$$Efficiency = \frac{OtherServicesEmployment}{RetailSquareFeet}$$

This estimates:

- how much service employment exists per retail inventory

Why interesting:

- oversupplied retail markets show low efficiency
- dense productive markets show high efficiency

Institutional retail models love inventory-normalized features.

5. Service Employment Volatility

Rolling instability matters more than average growth.

$$Volatility_t = StdDev(YoYGrowth_{last\ 8Q})$$

$$Volatility_t = StdDev(YoYGrowth_{last\ 8Q})$$

Why powerful:

- unstable service employment → unstable tenant demand
- vacancy persistence rises in volatile economies

Especially useful during:

- post-COVID normalization
- recession periods

6. Employment Recovery Gap

Measure distance from prior peak.

$$RecoveryGap = \frac{CurrentEmployment}{HistoricalPeakEmployment}$$

$$RecoveryGap = \frac{CurrentEmployment}{HistoricalPeakEmployment}$$

Why strong:

- many markets never fully recover after shocks
- vacancy behaves differently in partially recovered cities

Example:

- NYC office recovery
- SF retail recovery post-pandemic

7. Service Employment Pressure Index

Create labor pressure on local commercial space.

$$Pressure = \frac{EmploymentGrowth}{VacancyRate}$$

$$Pressure = \frac{EmploymentGrowth}{VacancyRate}$$

Interpretation:

- strong employment growth + low vacancy
→ leasing pressure building

This often predicts:

- rent growth
- future occupancy tightening

8. Local Consumer Activity Composite

Combine:

$$Composite = 0.4(ServiceEmploymentGrowth) + 0.3(PersonalIncomeGrowth) + 0.3(RetailSalesGrowth)$$

$$Composite = 0.4(ServiceEmploymentGrowth) + 0.3(PersonalIncomeGrowth) + 0.3(RetailSalesGrowth)$$

This creates:

- local demand pulse
- urban vitality score

Very realistic institutional-style feature.

9. Structural Dependency Ratio

$$DependencyRatio = \frac{OtherServicesEmployment}{TotalPrivateEmployment}$$

$$DependencyRatio = \frac{OtherServicesEmployment}{TotalPrivateEmployment}$$

This tells you:

- whether a market is overly dependent on small service jobs

High dependency markets:

- recover fast in expansions
- crash harder during downturns

This introduces cyclical sensitivity into vacancy forecasts.

10. Market Divergence Feature

Compare local market to national trend.

$$Divergence = LocalGrowth - NationalGrowth$$

$$Divergence = LocalGrowth - NationalGrowth$$

Very important.

Because investors care about:

- relative market strength
not just absolute growth.

A market outperforming national service employment trends often sees:

- stronger leasing
 - cap rate compression
 - lower vacancy
-

11. Service Employment Saturation

Detect mature markets.

$$Saturation = \frac{CurrentEmployment}{RollingTrend_{20Q}}$$

$$Saturation = \frac{CurrentEmployment}{RollingTrend_{20Q}}$$

Interpretation:

- far above trend → overheating
- below trend → underutilized economy

Can help predict:

- vacancy reversals
- mean reversion

12. Service-to-Office Interaction

For office vacancy models:

$$Interaction = OtherServicesGrowth \times OfficeEmploymentGrowth$$

$$Interaction = OtherServicesGrowth \times OfficeEmploymentGrowth$$

Reason:

- downtown ecosystems require both
- office alone is insufficient now post-COVID

This is extremely realistic.

13. Neighborhood Economic Fragility

$$Fragility = \frac{Volatility}{TrendGrowth}$$

$$Fragility = \frac{Volatility}{TrendGrowth}$$

High fragility markets:

- unstable leasing demand
- higher vacancy persistence
- weaker tenant quality

14. Employment Elasticity to Vacancy

Advanced panel feature:

Estimate:

$$Elasticity = \frac{\Delta Vacancy}{\Delta Employment}$$

$$Elasticity = \frac{\Delta Vacancy}{\Delta Employment}$$

Per market over rolling windows.

This captures:

- sensitivity of occupancy to service economy

Some markets are:

- highly employment-sensitive
- others structurally resilient

Huge alpha feature for institutional forecasting.

15. Hidden Recession Detector

$$RecessionSignal = 1(YoY Growth < 0 \wedge RollingMean_{4Q} < RollingMean_{12Q})$$

$$RecessionSignal = 1(YoY Growth < 0 \wedge RollingMean_{4Q} < RollingMean_{12Q})$$

This acts like:

- local economic deterioration detector

Very useful for:

- early vacancy stress prediction
- regime-switching models

What Top CRE Teams Usually Prefer

The most valuable features are usually:

Feature Type	Practical Power
Relative-to-national	Very high
Recovery gap	Very high
Regime states	Very high
Volatility	High
Interaction features	High
Density features	High
Simple lagged growth	Moderate

My Suggestion for Your Dataset

For quarterly panel real estate data across markets, I would prioritize:

1. Relative market divergence
2. Recovery gap
3. Volatility regime
4. Population-normalized density
5. Interaction with income growth
6. Service pressure index
7. Local fragility score
8. Market regime encoding

Those are much closer to what sophisticated CRE forecasting systems actually use.



Sorry I was asking can you look for some more top notch features engineering which top real estate industry use

Yes — if we think like top institutional CRE shops (Blackstone, Brookfield, Green Street, CBRE Econometric Advisors, Moody's Analytics CRE, Clarion, PGIM, etc.), they usually do **cross-market structural features**, **cycle-state signals**, and **spatial relative features**, not just basic time-series transformations.

For `Other_Services_Employment_Extended`, here are much more institutional-grade feature engineering ideas specifically for quarterly market-level vacancy forecasting.

1. Service Employment Beta to National Economy

Instead of raw growth, estimate:

$$\beta = \frac{\text{Cov}(\text{LocalGrowth}, \text{NationalGrowth})}{\text{Var}(\text{NationalGrowth})}$$

$$\beta = \frac{\text{Cov}(\text{LocalGrowth}, \text{NationalGrowth})}{\text{Var}(\text{NationalGrowth})}$$

This measures:

“How cyclically sensitive is this market’s service economy?”

Interpretation:

- high beta markets:
 - boom harder
 - crash harder
 - vacancy volatility higher

- low beta markets:
 - defensive markets
 - stable occupancy

This is heavily used in institutional portfolio allocation.

2. Employment Breadth Diffusion Signal

Instead of looking only at level growth:

Create:

$$Breadth = \frac{\#PositiveSubSectors}{\#TotalSubSectors}$$

$$Breadth = \frac{\#PositiveSubSectors}{\#TotalSubSectors}$$

Why elite:

- broad-based service growth is healthier
- narrow growth is fragile

Example:

- only salons growing = weak
- all service subsectors growing = strong ecosystem

This predicts:

- sustainability of occupancy recovery
-

3. Local Urban Vitality Factor

One of the most powerful modern CRE concepts post-COVID.

Build latent factor:

$$Vitality = f(\text{ServiceEmploymentGrowth}, \text{TransitMobility}, \text{RestaurantTraffic}, \text{RetailSales})$$

$$Vitality = f(\text{ServiceEmploymentGrowth}, \text{TransitMobility}, \text{RestaurantTraffic}, \text{RetailSales})$$

This captures:

- downtown activation
- mixed-use vibrancy
- neighborhood attractiveness

Especially predictive for:

- retail
- multifamily
- CBD office recovery

4. Service Employment Persistence Score

CRE investors care more about persistence than one-quarter spikes.

Estimate rolling autocorrelation:

$$Persistence = Corr(Growth_t, Growth_{t-1})$$

$$Persistence = Corr(Growth_t, Growth_{t-1})$$

Why useful:

- persistent employment trends drive leasing confidence
- unstable trends hurt tenant decision-making

5. Employment-to-Housing Imbalance

Very institutional.

$$Imbalance = ServiceEmploymentGrowth - HousingSupplyGrowth$$

$$Imbalance = ServiceEmploymentGrowth - HousingSupplyGrowth$$

Interpretation:

- employment growing faster than housing
 - tighter apartment markets
 - lower vacancy

Huge multifamily alpha feature.

6. Real Estate Demand Intensity Signal

Normalize employment by occupied space instead of total space.

$$DemandIntensity = \frac{OtherServicesEmployment}{OccupiedRetailSF}$$

$$DemandIntensity = \frac{OtherServicesEmployment}{OccupiedRetailSF}$$

This detects:

- tenant saturation
- leasing pressure

- underutilized retail ecosystems

Very sophisticated retail feature.

7. Service Employment Shock Absorption Capacity

Top firms model resilience.

$$Resilience = \frac{PostShockRecoverySpeed}{ShockMagnitude}$$

$$Resilience = \frac{PostShockRecoverySpeed}{ShockMagnitude}$$

Markets recovering rapidly after shocks:

- sustain occupancy better
- attract capital faster
- experience lower distress

8. Economic Complexity Proxy

Not obvious — but extremely powerful.

$$Complexity = 1 - \left(\frac{OtherServicesEmployment}{TotalEmployment} \right)$$

$$Complexity = 1 - \left(\frac{OtherServicesEmployment}{TotalEmployment} \right)$$

Idea:

- overly service-dependent markets may lack diversified economic engines
- diversified markets have more resilient vacancy dynamics

This becomes powerful in downturns.

9. Labor Quality Migration Signal

Use interaction:

$$MigrationSignal = ServiceEmploymentGrowth \times WageGrowth$$

$$MigrationSignal = ServiceEmploymentGrowth \times WageGrowth$$

Why elite:

- job growth without wage growth may indicate weak labor quality
- high wage + service growth indicates affluent migration markets

Very predictive for:

- Sunbelt multifamily
- mixed-use demand

10. Spatial Competitive Advantage

Institutional firms compare markets regionally.

$$RelativeStrength = LocalGrowth - RegionalAverageGrowth$$

$$RelativeStrength = LocalGrowth - RegionalAverageGrowth$$

Capital flows toward:

- relative winners
not absolute winners.

This matters enormously for:

- cap rates
- leasing momentum
- vacancy compression

11. Service Employment Cycle Position

Estimate cyclical percentile:

$$CyclePosition = PercentileRank(CurrentEmployment, History)$$

$$CyclePosition = PercentileRank(CurrentEmployment, History)$$

Why important:

- late-cycle markets behave differently
- vacancy mean reversion increases at extremes

This is heavily used in tactical allocation.

12. Tenant Formation Probability

Small businesses create tenancy demand.

Estimate:

$$FormationProxy = \Delta OtherServicesEmployment \times SmallBusinessGrowth$$

$$FormationProxy = \Delta OtherServicesEmployment \times SmallBusinessGrowth$$

This predicts:

- strip retail absorption
- neighborhood leasing demand

Very realistic.

13. Service Employment Convexity

Institutional models increasingly use nonlinear dynamics.

$$Convexity = Growth_t - 2Growth_{t-1} + Growth_{t-2}$$

$$Convexity = Growth_t - 2Growth_{t-1} + Growth_{t-2}$$

This captures:

- acceleration curvature
- inflection points
- regime transitions

Excellent for early turning-point detection.

14. Vacancy Sensitivity Regime

Create market-specific sensitivity score.

$$Sensitivity = Corr(ServiceGrowth, VacancyChange)$$

$$Sensitivity = Corr(ServiceGrowth, VacancyChange)$$

Some markets:

- highly reactive to local services
- Others:
- driven by finance/tech/manufacturing

This becomes a meta-feature across markets.

15. Economic Recovery Synchronization

Very advanced institutional feature.

$$Synchronization = Corr(LocalServiceGrowth, NationalCRECycle)$$

$$Synchronization = Corr(LocalServiceGrowth, NationalCRECycle)$$

Markets highly synchronized with national CRE cycles:

- reprice faster
- vacancy deteriorates faster during downturns

Low synchronization:

- diversification benefits

The Features That Actually Create Alpha

In real institutional forecasting, the biggest gains usually come from:

Feature Class	Importance
Relative market strength	Extremely high
Regime/state features	Extremely high
Recovery/resilience metrics	Extremely high
Interaction terms	Very high
Cross-market normalization	Very high
Spatial features	Very high
Simple lag growth	Low

What I Would Personally Build First

For your quarterly panel dataset:

Tier 1

- RelativeStrength
- RecoveryGap
- Persistence
- CyclePosition
- VolatilityRegime

Tier 2

- MigrationSignal
- VitalityFactor
- Resilience
- Imbalance
- DemandIntensity

Tier 3

- Convexity
- BetaToNational
- Synchronization
- Fragility
- BreadthDiffusion

These are much closer to what sophisticated CRE macro forecasting systems actually use internally.



Show me some more features engineering with `Other_Services_Employment_Extended` which are realistic and don't try to keep the earlier feature.

Absolutely — let's move into a different class of feature engineering entirely and avoid repeating earlier concepts.

For `Other_Services_Employment_Extended`, sophisticated CRE teams often think about:

- labor market structure
- hidden tenant demand
- elasticity of local spending
- service-sector saturation
- macro transmission mechanisms
- occupancy friction
- cyclical asymmetry

Below are fresh, realistic, institutional-style feature ideas that are materially different from the previous ones.

1. Service Employment Per Household

$$ServicePerHousehold = \frac{OtherServicesEmployment}{Households}$$

$$ServicePerHousehold = \frac{OtherServicesEmployment}{Households}$$

Why important:

- measures local service penetration
- stronger proxy for neighborhood commercial intensity than raw population

Higher values often indicate:

- denser consumption ecosystems

- stronger strip retail utilization
- lower neighborhood retail vacancy

2. Employment Utilization Gap

Measure deviation from expected employment.

First estimate trend:

$$Expected_t = RollingTrend(Employment)$$

Then:

$$Gap_t = \frac{Employment_t - Expected_t}{Expected_t}$$

$$Gap_t = \frac{Employment_t - Expected_t}{Expected_t}$$

Why useful:

- detects hidden overheating or underperformance
- vacancy often responds to deviations from equilibrium, not levels

3. Service Employment Compression Indicator

Measure whether growth is slowing despite still-positive levels.

$$Compression = \frac{Growth_t}{Growth_{t-4}}$$

$$Compression = \frac{Growth_t}{Growth_{t-4}}$$

Interpretation:

- below 1 → slowing expansion
- early warning before vacancy deterioration

Very realistic for cyclical turning points.

4. Local Consumption Support Ratio

$$ConsumptionSupport = \frac{OtherServicesEmployment}{RetailVacancy}$$

$$ConsumptionSupport = \frac{OtherServicesEmployment}{RetailVacancy}$$

Why strong:

- combines labor demand with retail stress

- higher ratio indicates healthier tenant ecosystem

Especially useful in:

- neighborhood retail
- mixed-use corridors

5. Employment-to-New-Supply Ratio

$$SupplyAbsorptionCapacity = \frac{OtherServicesEmploymentGrowth}{NewSupplyGrowth}$$

$$SupplyAbsorptionCapacity = \frac{OtherServicesEmploymentGrowth}{NewSupplyGrowth}$$

Institutional relevance:

- determines whether local demand can absorb incoming inventory
- critical for forecasting vacancy spikes

Very important in Sunbelt markets.

6. Service Employment Duration Above Trend

Instead of level, measure persistence.

$$Duration = ConsecutiveQuarters(Employment > Trend)$$

Why powerful:

- prolonged above-trend growth often precedes rent acceleration
- sustained economic strength matters more than temporary spikes

CRE markets react to duration.

7. Economic Friction Indicator

$$Friction = \frac{|QoQGrowth - YoYGrowth|}{|YoYGrowth|}$$

$$Friction = \frac{|QoQGrowth - YoYGrowth|}{|YoYGrowth|}$$

Interpretation:

- high friction = unstable growth structure
- indicates transitional economic conditions

Vacancy tends to rise when labor trends become inconsistent.

8. Employment Concentration Drift

Measure whether services are becoming disproportionately dominant.

$$Drift = Share_t - Share_{t-8}$$

Where:

$$Share = \frac{OtherServicesEmployment}{TotalEmployment}$$

$$Share = \frac{OtherServicesEmployment}{TotalEmployment}$$

Why sophisticated:

- captures structural transformation of local economies
- excessive drift may imply lower-quality economic expansion

9. Service Employment Mean-Reversion Pressure

$$Pressure = \frac{Employment_t - RollingMean_{20Q}}{RollingStd_{20Q}}$$

$$Pressure = \frac{Employment_t - RollingMean_{20Q}}{RollingStd_{20Q}}$$

This is essentially a macro z-score.

Why useful:

- extreme positive deviations often precede cooling
- extreme negative deviations often precede rebound

Very common in institutional cycle models.

10. Tenant Ecosystem Stability

$$Stability = 1 - \frac{QuarterlyLayoffProxy}{OtherServicesEmployment}$$

$$Stability = 1 - \frac{QuarterlyLayoffProxy}{OtherServicesEmployment}$$

Even approximate layoff proxies help.

Why useful:

- stable service labor base supports occupancy durability
- unstable labor → tenant churn risk

11. Economic Spillover Multiplier

Measure spillover into real estate demand.

$$Spillover = \frac{\Delta OccupiedSF}{\Delta OtherServicesEmployment}$$

$$Spillover = \frac{\Delta OccupiedSF}{\Delta OtherServicesEmployment}$$

Interpretation:

- how efficiently employment growth converts into occupancy demand

Some markets generate:

- high employment
but weak real estate absorption.

This feature captures that disconnect.

12. Service Employment Exhaustion Signal

$$Exhaustion = RollingMax_{12Q}(Growth) - CurrentGrowth$$

$$Exhaustion = RollingMax_{12Q}(Growth) - CurrentGrowth$$

Why elite:

- identifies late-cycle deceleration
- useful before vacancy inflection

13. Employment-to-Transit Dependency

$$TransitDependency = \frac{OtherServicesEmployment}{TransitRidership}$$

$$TransitDependency = \frac{OtherServicesEmployment}{TransitRidership}$$

Powerful in:

- urban office
- mixed-use districts
- CBD retail

Captures whether local service ecosystem depends heavily on commuter flows.

14. Local Service Labor Tightness

$$Tightness = \frac{JobOpenings}{OtherServicesEmployment}$$

$$Tightness = \frac{JobOpenings}{OtherServicesEmployment}$$

Why realistic:

- labor shortages can constrain business expansion
- affects future leasing demand

Especially relevant post-2021.

15. Demand Conversion Efficiency

$$ConversionEfficiency = \frac{NetAbsorption}{EmploymentGrowth}$$

$$ConversionEfficiency = \frac{NetAbsorption}{EmploymentGrowth}$$

This measures:

- how effectively local economic growth converts into occupancy

Institutional investors care deeply about this.

Some markets:

- generate jobs but little occupancy
- Others:
- efficiently translate growth into leasing demand

16. Employment Regime Transition Velocity

$$Velocity = RegimeScore_t - RegimeScore_{t-1}$$

Why advanced:

- markets shifting rapidly between regimes have unstable vacancy dynamics
- transition speed matters more than regime alone

17. Service Labor Saturation Risk

$$SaturationRisk = \frac{OtherServicesEmploymentGrowth}{PopulationGrowth}$$

$$SaturationRisk = \frac{OtherServicesEmploymentGrowth}{PopulationGrowth}$$

Interpretation:

- employment growing much faster than population
 - possible oversaturation
 - lower long-term sustainability

18. Occupancy Buffer Capacity

$$Buffer = OccupiedInventoryGrowth - EmploymentGrowth$$

$$Buffer = OccupiedInventoryGrowth - EmploymentGrowth$$

Useful because:

- markets with shrinking occupancy buffer are vulnerable to vacancy spikes

19. Service Economy Inefficiency Signal

$$Inefficiency = \frac{EmploymentGrowth}{IncomeGrowth}$$

$$Inefficiency = \frac{EmploymentGrowth}{IncomeGrowth}$$

Interpretation:

- job growth without income growth may imply weak-quality expansion

Can predict:

- fragile tenant demand
- weaker rent growth

20. Local Economic Stickiness

$$Stickiness = 1 - TurnoverRate$$

or approximate using employment stability.

Why powerful:

- sticky labor markets create sticky occupancy
- transient labor markets increase vacancy volatility

Very important for multifamily forecasting.